NANYANG TECHNOLOGICAL UNIVERSITY
SCHOOL OF ELECTRICAL & ELECTRONIC ENGINEERING
ACADEMIC YEAR 2024/25 (SEMESTER 1)

IE2104/IM1004 DIGITAL ELECTRONICS
TUTORIAL 8

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1. Describe what will happen to the circuit shown in Fig. 1 when A which is initially logic "0" switches to logic "1" after 10ns. Assume all the gates have the same delay of 5ns. Sketch the waveforms for P, Q, R and X and determine the frequency and duty cycle of signal X.

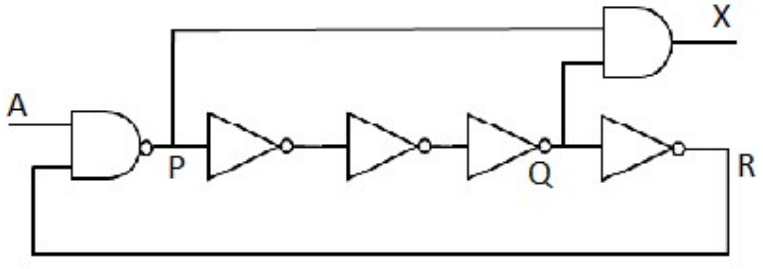
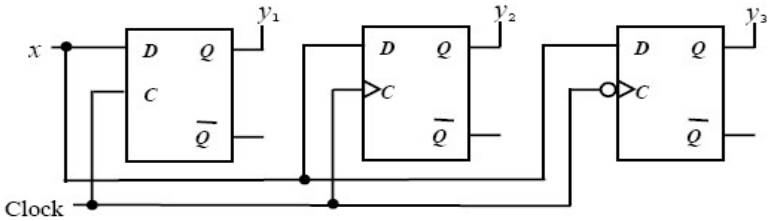


Fig. 1.



2. Draw the two basic latch configurations and write the characteristic function table and excitation table.

3. The 3 flip-flops in the circuit below in Fig. 2 have different triggering modes. Given the input *x* as shown and the clock signal, draw the waveforms for the outputs for the 3 flip-flops.



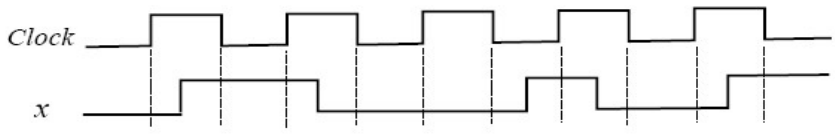


Fig. 2

4. Clocked *SR* flip-flop can be constructed from cross-coupled NOR gate as shown in Fig. 3(a).

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Derive the characteristic table, characteristic equation and excitation table of the clocked SR flip-flop. Sketch the output waveform of *Q* for the given input waveforms in Fig. 3(b), if the propagation delay of AND and NOR gates are 7ns and 5ns respectively.

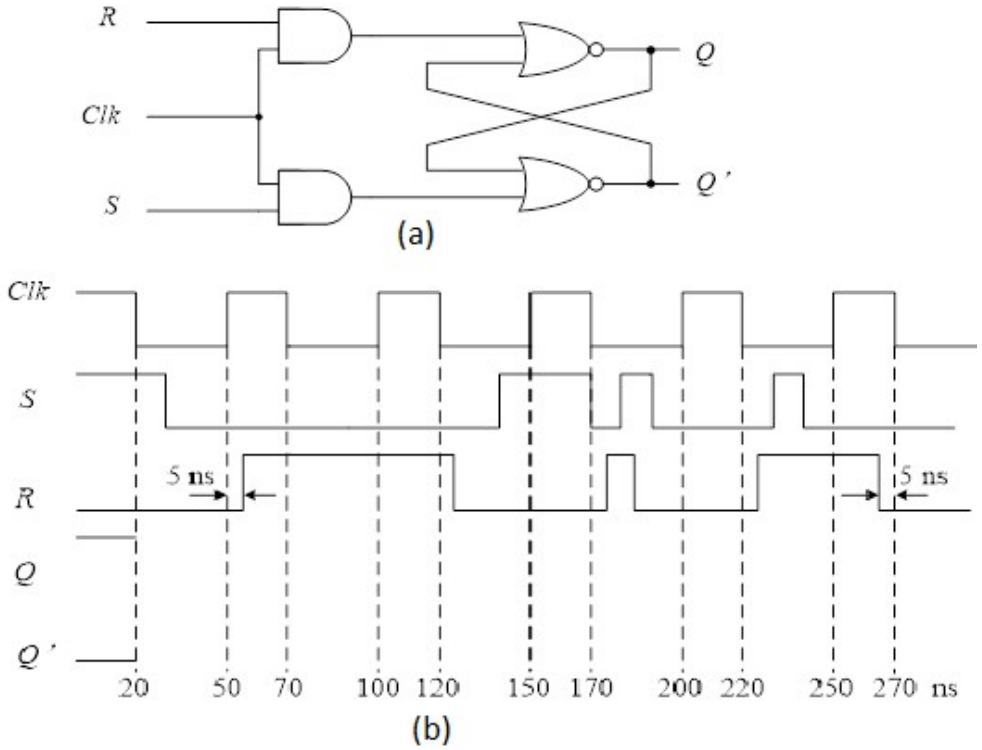


Figure 3.

5. A flip-flop which is constructed from a D flip-flop is as given in Fig. 4

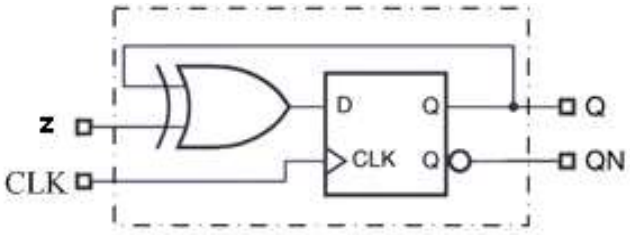


Fig. 4

Write the characteristic table, characteristic equation and excitation table of the flip-flop and specify the equivalent flip-flop as result of the additional logic gate.

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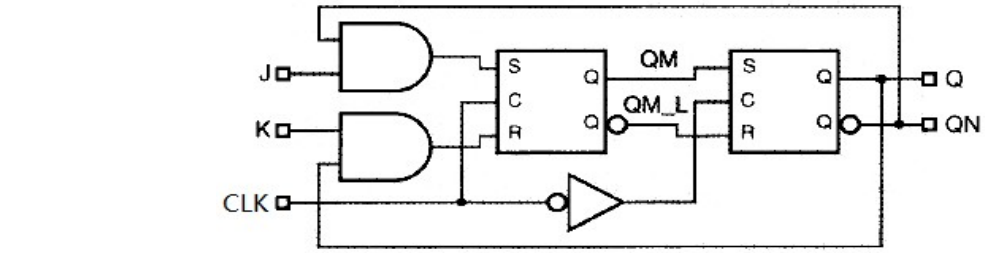
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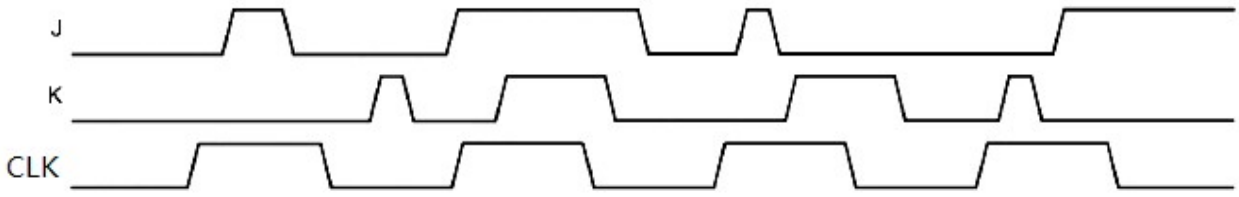
1. Starting with a basic NOR cell, design a positive level-triggered SR flip-flop.

2. A dual-mode flip-flop has two inputs, M and X. M is the mode input which causes the device to operate either as a D flip-flop or T flip-flop. When M is 0, the device behaves like a D flip-flop with X serving as the D input. When M is 1, the device behaves like a T flip-flop with X serving as the T input. Design the flip-flop using NAND cell.

3. This exercise shows how a level triggered flip-flop can become unstable under certain input conditions.
   a. For the SR flip-flop in Question 1, convert it to JK flip-flop by adding the necessary decoding logic. Merge the decoding logic with the logic gates of the SR flip-flop.
   b. Initially, the flip-flop is at a reset state with J and K at zero. When the clock goes high, J and K are simultaneously brought to logic "1" for a period ten times longer than the propagation delay of a logic gate. Show the effects of this on the Q and Q′ outputs with the aid of a timing diagram. Assume that all the gates in the circuit have the same propagation delay.

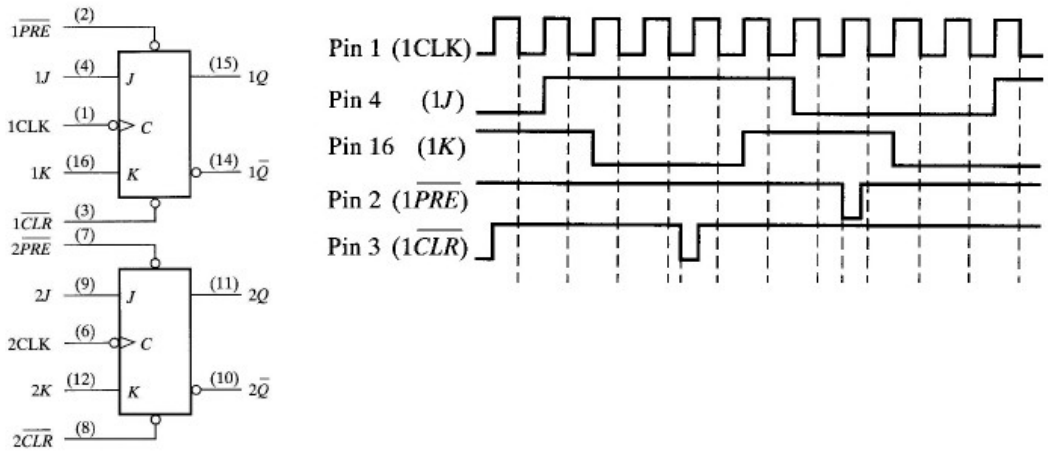
4. This exercise illustrates the problem of one's and zero's catching inherent in master-slave JK flip-flops.
   a. In the given timing diagram, fill in the following signals: QM, QM_L, Q and QN. Both the master and slave flip-flops are initially in reset states.
   b. Notice that the signals at Q and QN at the falling edge of the clock do not necessarily conform to the characteristic table of the JK flip-flop. Give an explanation for this observation.
   c. Suggest how this problem can be overcome.





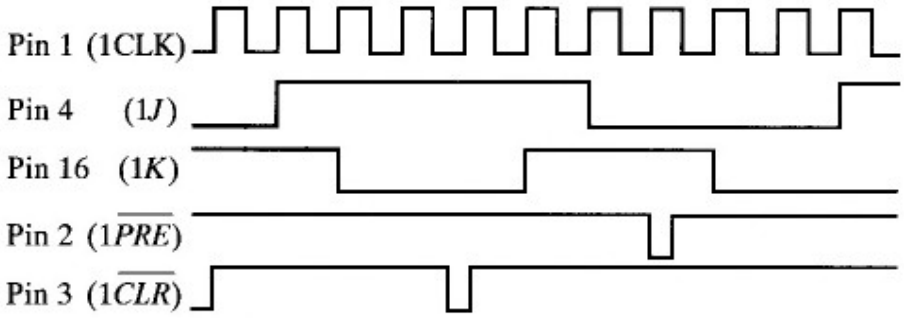
5. Shown below is a 74LS76A, a commercially available chip. Complete the accompanying timing waveforms by sketching the waveform for output 1Q (Pin 15).

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Additional practice questions (no solutions given).

I. Assume that the chip in Q5 is positive edge-triggered. Draw the corresponding waveform for output 1Q (Pin 15).



II. Starting with a basic NOR cell, design a negative level-triggered SR flip-flop.

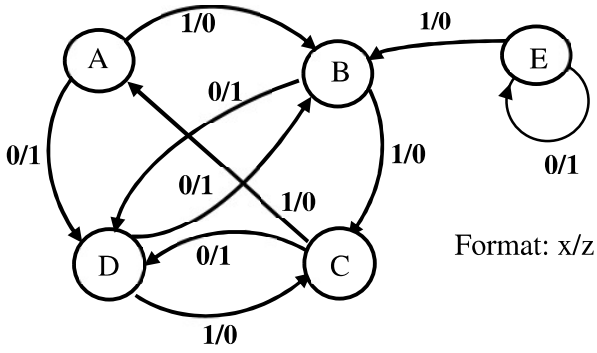
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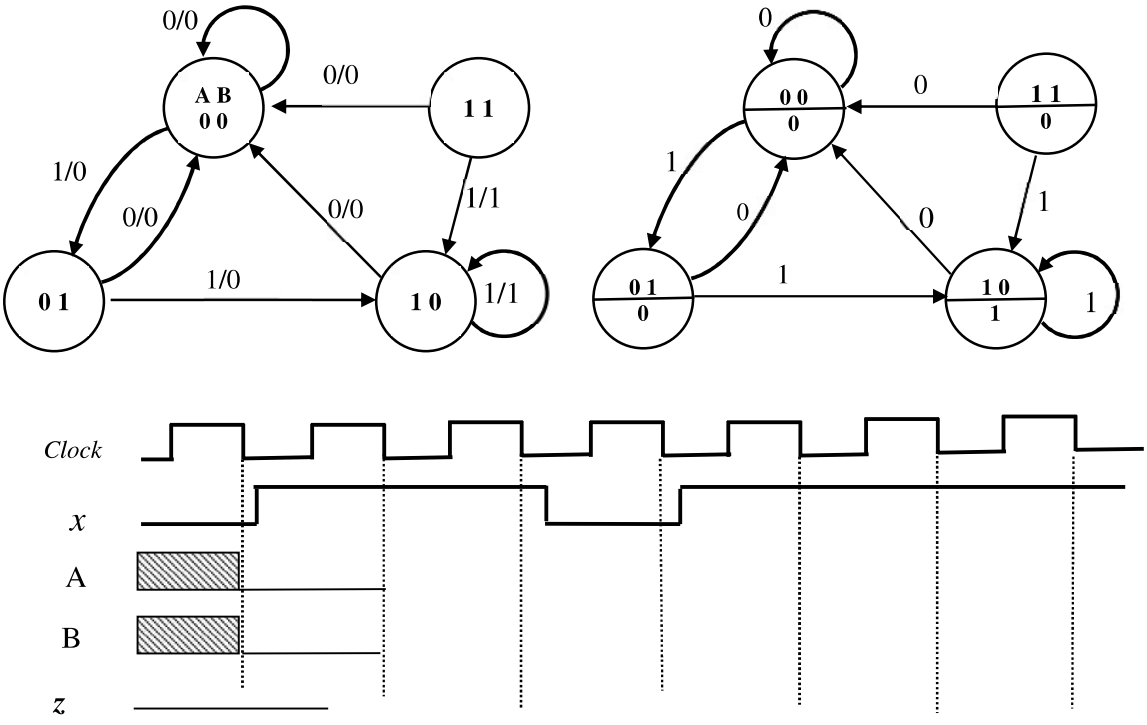
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1. Derive the state table for the following state diagram. If the system starts at state A, analyze the diagram to determine which state is not reachable under normal operation.



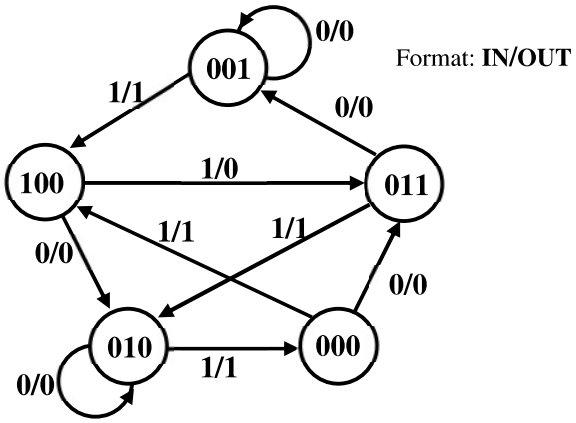
2. Complete the waveforms for A, B and z for the x input shown for both state machines. Assume falling edge triggered flip-flops are used for the state variables.



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3. Analyze and draw the state diagram for the circuit shown below.

4. A sequential circuit as shown by the state diagram shown below has one input IN and one output OUT. Design the circuit using T flip-flops. Analyze the true next states for the unused states.

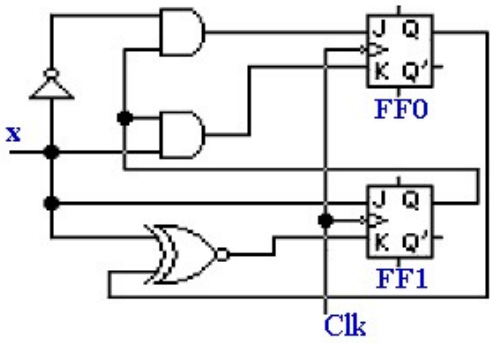


5. Design a Moore state machine which has an input *x* and an output *z*. The output z is initialized to 0. The output *z* = 1 if two consecutive '0's, or two consecutive '1's have been received at the input *x*. When either condition occurs, the output remains 1 until reset by an external reset signal. Draw the state diagram and show the state table. Assign 00 to the initial state, implement the state machine using JK flip-flops and the next state decoder using only two-input NAND gates.

Additional self-practice questions (no solutions given).

I. Draw a Moore state diagram that generates a 25% duty cycle periodic output. Using T flip-flops, draw the circuit.

II. Draw the state diagram for the circuit shown below.



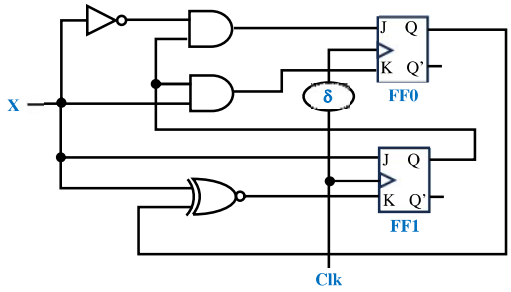
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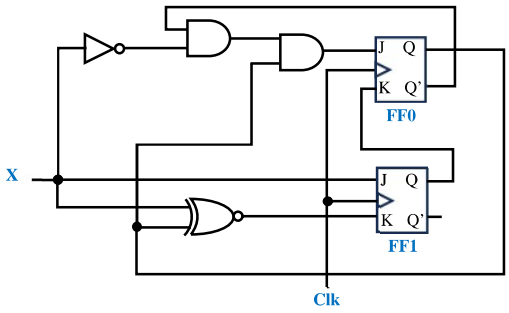
1. For the circuit shown below with device timing parameters given in the table, determine the setup time margin and hold time margin if the frequency of *Clk* is 20 MHz and there is no clock skew between FF0 and FF1, i.e., δ = 0. What is the maximum clock frequency for the circuit to operate without setup and hold time violation? What will be the setup time margin and hold time margin if δ = 3 ns?



Device	t_d (min)	t_d (max)	t_{su} (setup)	t_h (hold)
INV	3	4	-	-
AND	4	7	-	-
EXNOR	6	10	-	-
JK Flip-flop	4	12	8	5

Note: The timing values shown are in nanoseconds.

2. Consider the circuit. Using the same device timing parameters as given in the table in Q1, repeat the analysis as specified in Q1.



3. Draw the configurations of 4-bit asynchronous and synchronous straight sequence counters.

4. Implement an asynchronous counter with a straight sequence of 0000 through 1011.

5. Assume a binary octal representation of a counting sequence. Design a synchronous counter that counts in the sequence 0, 1, 5, 2, 6, 0,… using:
   a) T flip-flops
   b) D flip-flops

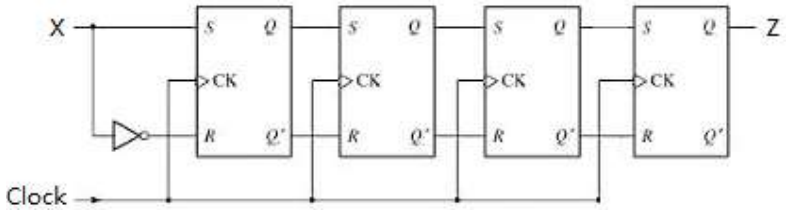
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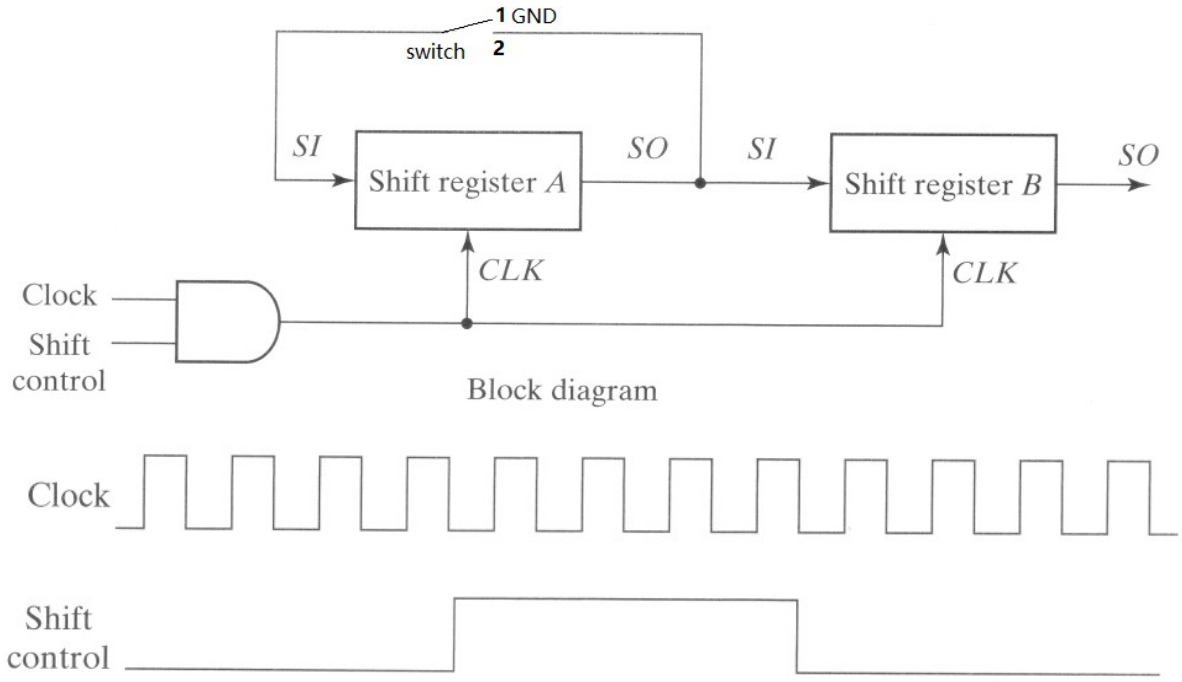
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1. Analyze and describe the function of the circuit below and draw the state diagram.



2. Study the circuit below and then sketch the waveforms for A[3..0] and B[3..0] with the switch in position 1 (to GND). Assume that both registers are triggered by the rising edge of the clock signal and the initial values in registers A and B are 1101 and 0000 respectively. Repeat with the switch in position 2 (connected).

   If T_{AB} is the difference in time between the arrivals of the clock rising edge at registers *A* and *B*, and T_{dff} is the minimum propagation delay of the flip-flop in the registers, describe what happens if the values of T_{AB} and T_{dff} are 5 and 7 ns respectively. What if T_{AB} is 9 ns instead?



3. How many address lines and data lines are needed for the following memory modules?
   a) 16K × 8 (b) 256K × 64 (c) 2G × 16 (d) 20 × 4 (e) 16 Mbytes

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4. For an 8 Kbytes SRAM, what is the total number of cells? How many address lines are required? For arrangement as a square array of cells in a single IC chip or module, show an example of a possible physical layout?

5. You are given a Hitachi 256Mbit DRAM. Each memory cell needs to be refreshed every 64ms.
    a. How many input pins are required to cater for the address lines?
    b. How is the memory configured in terms of number of rows and columns?
    c. How many refresh cycles are required for the whole chip?
    d. What is the refresh period for each cell?
    e. If the refresh time is 35ns, how many refresh cycles can be accommodated within one refresh period?
    f. What percentage of time is expended on refreshing?

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